Enzymes & pH

Exam code: 0610 & 0970

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Written by: Lára Marie McIvor
Reviewed by: Lucy Kirkham
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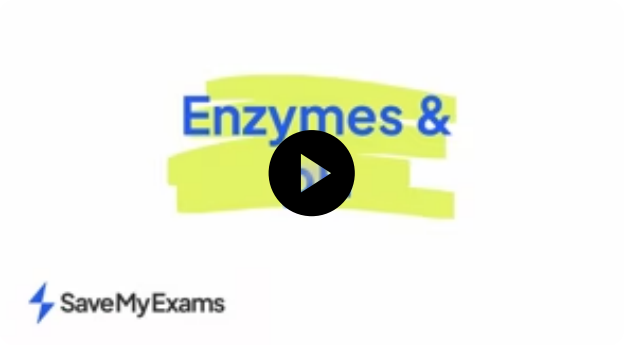

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Enzymes & pH: Extended

Extended Tier Only

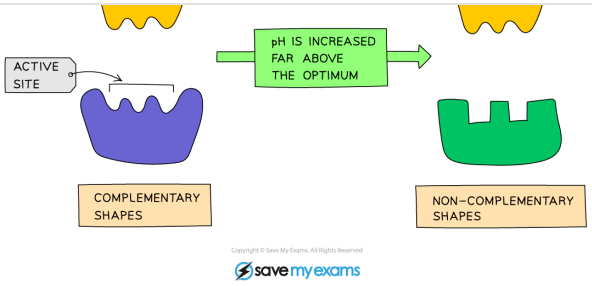
- The optimum pH for most enzymes is 7 but some that are produced in acidic conditions, such as the stomach, have a lower optimum pH (pH 2) and some that are produced in alkaline conditions, such as the duodenum, have a higher optimum pH (pH 8 or 9)
- If the pH is too high or too low, the bonds that hold the amino acid chain together to make up the protein can be destroyed
- This will change the shape of the active site, so the substrate can no longer fit into it, reducing the rate of activity
- Moving too far away from the optimum pH will cause the enzyme to denature and activity will stop

Help me understand

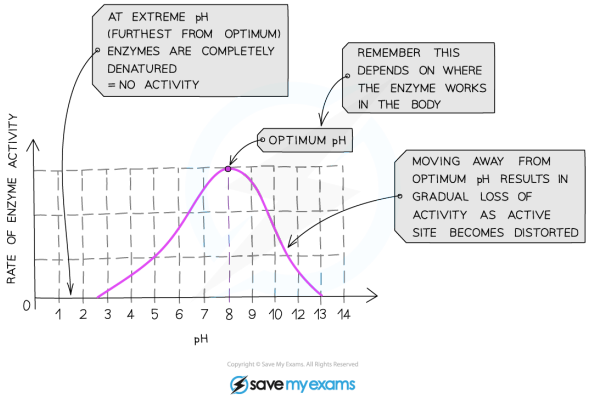
Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



Effect of pH on enzyme activity



Graph showing the effect of pH on rate of activity for an enzyme from the duodenum

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